

Multivariate Meta-Analysis for Longevity and Reproduction

Fay Frost, Daniel Noble, & Liam Dougherty

This documents the exploratory multivariate meta-analysis for the effects of temperature on the longevity and reproduction of species.

Multivariate meta-analysis and meta-regression

Reproductive traits were more strongly impacted by temperature changes than longevity when controlling for temperature (contrast = -1.04, 95% CI: -1.31 to -0.76, $df = 396$, p -value = < 0.0001). At 25°C, longevity was reduced by -0.22 SD units relative to the control (95% CI: -0.41 to -0.04, $df = 396$, p -value = 0.02). In contrast, reproduction was significantly reduced by -1.26 SD units relative to the control (95% CI: -1.52 to -1, $df = 396$, p -value = < 0.0001). However, there was high heterogeneity in both traits with prediction intervals spanning -6.35 to 4.12 (Figure ??).

Mean effects changed in complicated ways with changes in temperature, with significant non-linear patterns that depended on the effect outcome (Table ??), with effect magnitude and direction changing non-linearly with temperature increases. Interestingly, studies that observed larger changes in reproduction also observed correlated changes in longevity (between study correlation = 0.37), however, at the within study-level there was a much weaker correlation (within-study correlation = 0.17, Figure ??).

Table 1: Non-linear effects of temperature on mean effect size for longevity and reproduction

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Longevity (mean)	-0.0903	-0.3092	0.1286	392	0.42
Reproduction (mean)	-0.4837	-0.7943	-0.1732	392	< 0.01
Linear Treatment Temperature (centered) - Longevity	-0.2581	-0.2875	-0.2288	1,576	< 0.0001
Linear Treatment Temperature (centered) - Reproduction	-0.0660	-0.0978	-0.0342	1,576	< 0.0001

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Quadratic Treatment Temperature (centered) - Longevity	-0.0022	-0.0042	-0.0002	1,576	0.03
Quadratic Treatment Temperature (centered) - Reproduction	-0.0123	-0.0146	-0.0099	1,576	< 0.0001
Cubic Treatment Temperature (centered) - Longevity	0.0005	0.0003	0.0006	1,576	< 0.0001
Cubic Treatment Temperature (centered) - Reproduction	-0.0001	-0.0002	0.0001	1,576	0.42

Sensitivity Analysis

In this analysis we remove any effect sizes above 10 or below -10. When removing the 10% largest and smallest effect sizes, the following effect sizes are removed.

```
[1] TEMP559_A.-10 TEMP1964A.5    TEMP1964A.7    TEMP2281A.10  TEMP546_A.10
[6] TEMP559_A.15
21976 Levels: HUM109A.-26 HUM148A.-26 HUM22B.-26 HUM256A.-26 ... TEMP995A.21
```

Splitting analysis: warm versus cool

In this analysis we split the dataset into studies warmer and cooler than the reference temperature of 25 degrees C. Here we explored non-linear effects up to degree 4. We found that there was a significant and negative linear effect of temperature on both reproduction and adult lifespan at warmer temperatures (-0.17 and -0.38 respectively, (Table Table ??). However, reproductive traits were more strongly impacted by increased temperature changes than adult lifespan (contrast = -0.21, 95% CI: -0.26 to -0.16, df = 786, *t*-value = -8.59, *p*-value = < 0.0001). Figure Figure ?? models models shows the predicted linear effects of temperature on reproduction and adult lifespan for observed treatment temperatures warmer than their reference temperature.

Interestingly, Figure Figure ?? shows that studies which observed larger changes in reproduction also observed correlated changes in longevity (between study correlation = 0.65, 95% CI: 0.52 to 0.75), however, at the within study-level there was a much weaker correlation (within-study correlation = 0.5, 95% CI: 0.38 to 0.61)

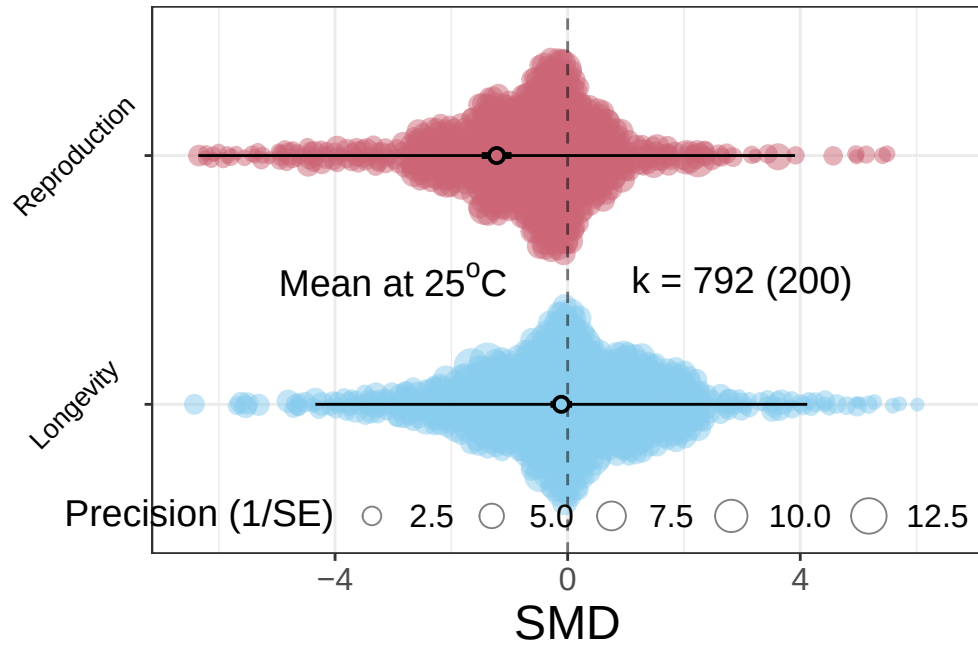


Figure 1: Orchard plot with the overall meta-analytic mean (central dots) for reproduction and longevity traits. Thick bars are the 95 confidence intervals and thin bars are the 95 prediction intervals. Data are weighted by their precision (inverse sampling error). Plot is truncated for ease of visualisation. k is the number of effect sizes and n the number of studies. Given we limited our analysis to studies with both reproduction and longevity the samples sizes are the same for both trait types.

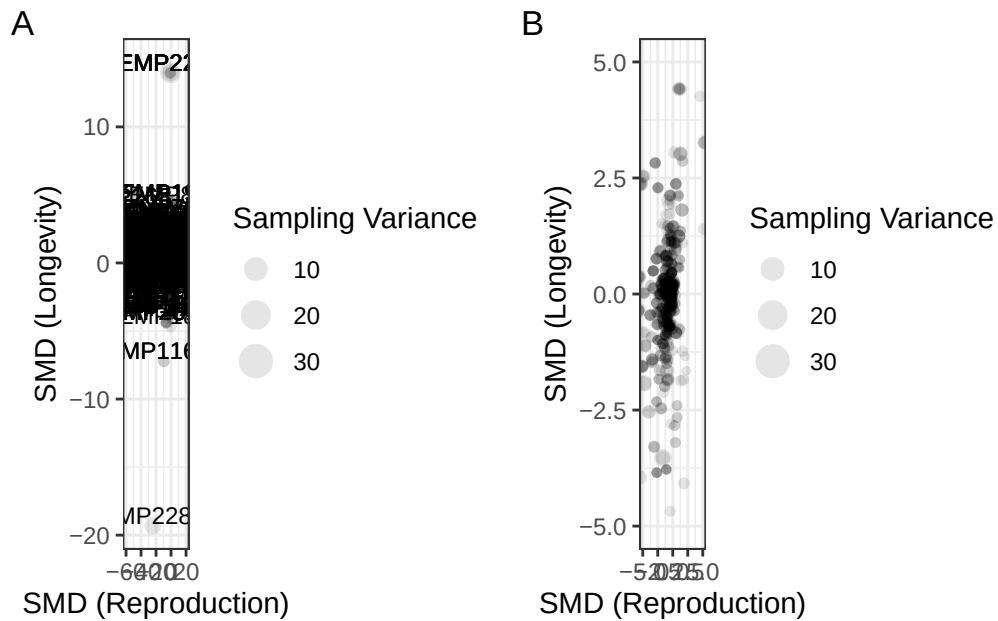


Figure 2

Table 2

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Longevity (mean)	-0.53	-0.89	-0.18	344	< 0.01
Reproduction (mean)	0.54	0.01	1.08	344	0.05
Linear Treatment Temperature (centered) - Longevity	-0.17	-0.21	-0.13	786	< 0.0001
Linear Treatment Temperature (centered) - Reproduction	-0.38	-0.44	-0.32	786	< 0.0001

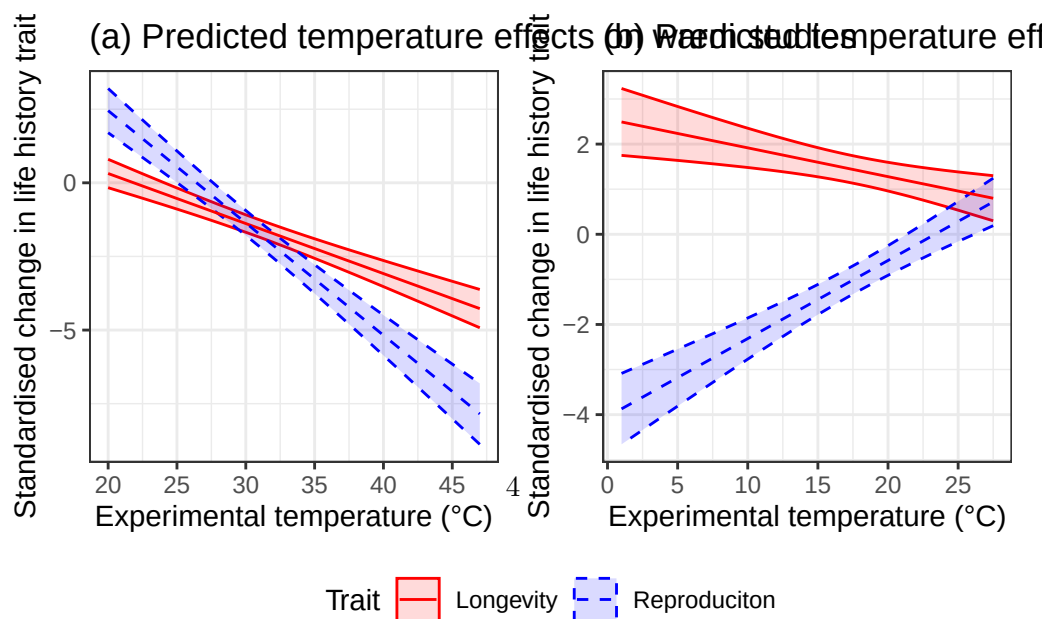


Figure 3

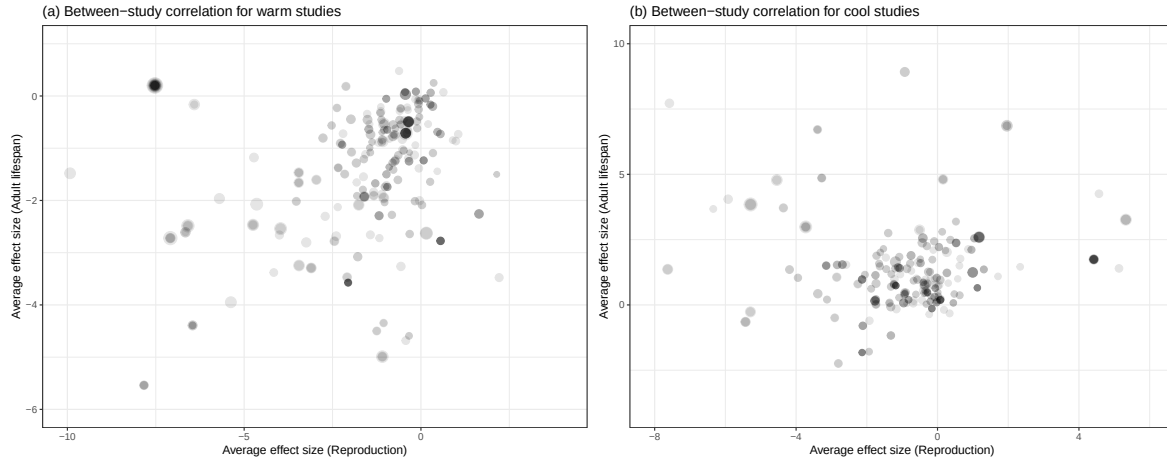


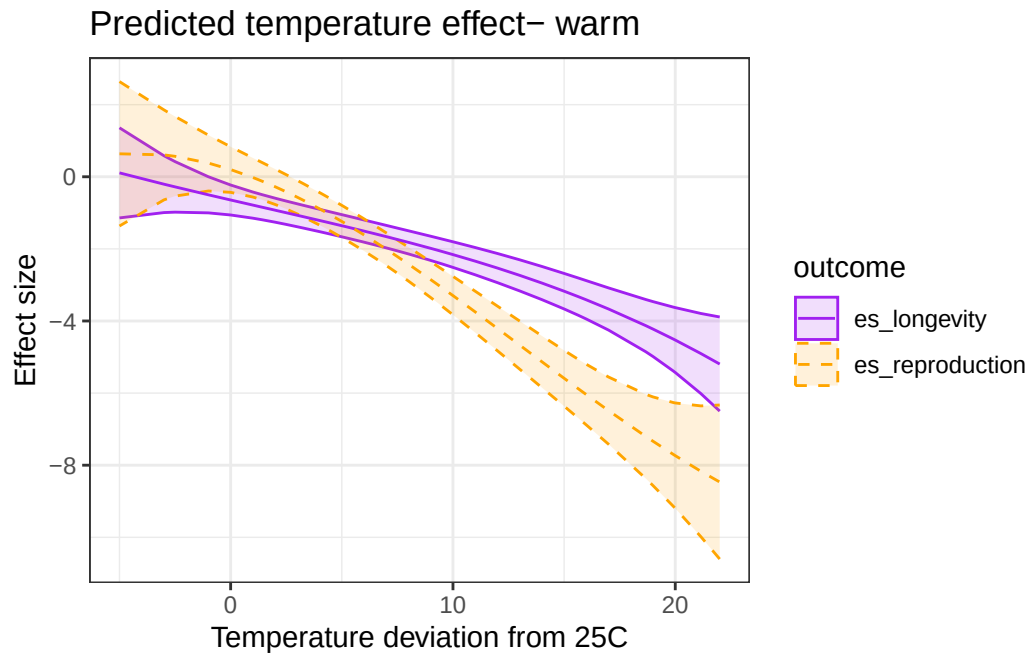
Figure 4

temperature and reproduction (see Table 4 for estimates). Figure 7 shows the predicted relationship between temperature and reproduction and adult lifespan for observed treatment temperatures cooler than their reference temperature.

We found that studies that observed larger changes in reproduction also observed correlated changes in longevity (between study correlation = -0.15, 95% CI: -0.31 to 0.09), however, at the within study-level there was a much weaker correlation (within-study correlation = -0.23, 95% CI: -0.36 to -0.09)

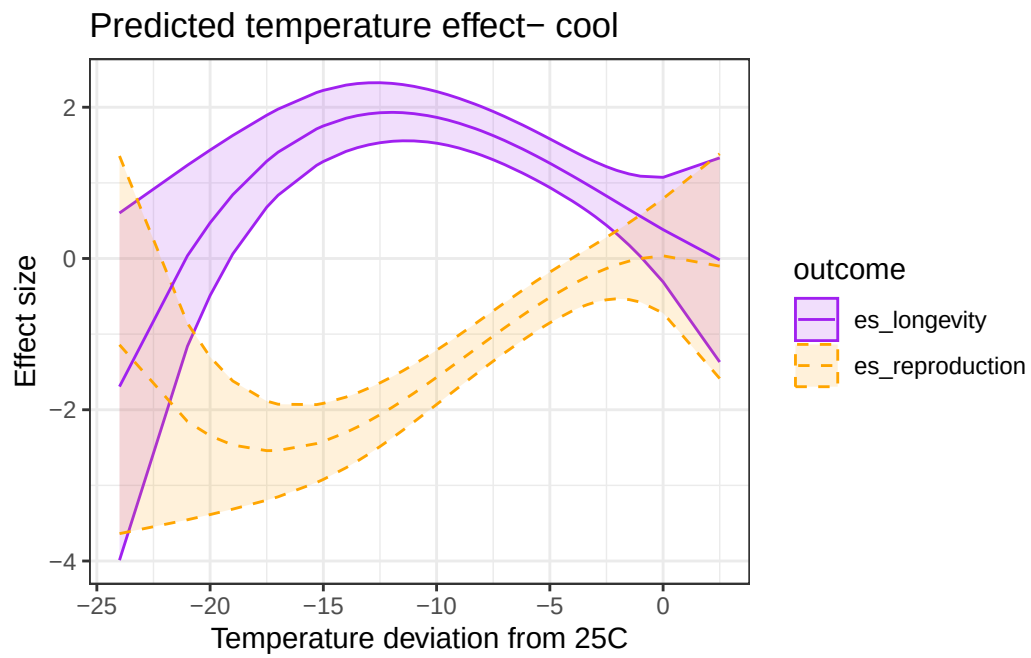
Table 3

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Longevity (mean)	0.0972	-0.4646	0.6590	320	0.73
Reproduction (mean)	0.4945	-0.1196	1.1087	320	0.11
Linear Treatment Temperature (centered) - Longevity	-0.3271	-0.3955	-0.2586	790	< 0.0001
Linear Treatment Temperature (centered) - Reproduction	0.1484	0.0754	0.2214	790	< 0.001
Quadratic Treatment Temperature (centered) - Longevity	-0.0146	-0.0191	-0.0101	790	< 0.0001
Quadratic Treatment Temperature (centered) - Reproduction	-0.0003	-0.0052	0.0046	790	0.91



prediction intervals for mv_mlma split by warm.cool- cubic models-1}

{#fig-



prediction intervals for mv_mlma split by warm.cool- cubic models-2}

{#fig-

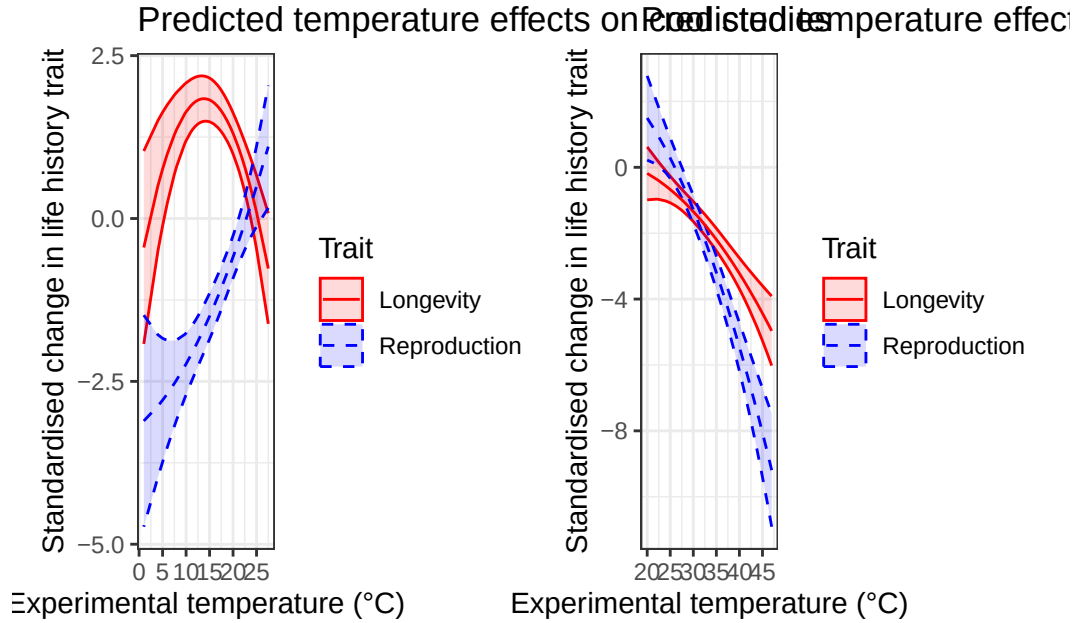


Figure 5

Sensitivity analysis- Robust variance estimation

Sensitivity analysis- removing outliers

Warm subset

Table 4: Linear effects of temperature on mean effect size for longevity and reproduction when treatment temperatures are warmer than reference temperatures (sensitivity analysis with 10% outliers removed).

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Longevity (mean)	-0.59	-0.852	-0.327	337	< 0.0001
Reproduction (mean)	0.30	-0.054	0.660	337	0.1
Linear Treatment Temperature (centered) - Longevity	-0.13	-0.160	-0.097	759	< 0.0001
Linear Treatment Temperature (centered) - Reproduction	-0.26	-0.305	-0.216	759	< 0.0001

We found weaker, but qualitatively similar correlations, after removing outliers (between-study correlation = 0.47, 95% CI: 0.27 to 0.63 within-study correlation = 0.46, 95% CI: 0.32 to 0.59)

Cool subset

Table 5: quadratic effects of temperature on mean effect size for longevity and reproduction when treatment temperatures are cooler than reference temperatures (sensitivity analysis with 10% outliers removed).

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Longevity (mean)	0.3407	-0.1321	0.813	317	0.16
Reproduction (mean)	0.3858	-0.1916	0.963	317	0.19
Linear Treatment Temperature (centered) - Longevity	-0.2892	-0.3448	-0.234	773	< 0.0001
Linear Treatment Temperature (centered) - Reproduction	0.1403	0.0729	0.208	773	< 0.0001
Quadratic Treatment Temperature (centered) - Longevity	-0.0139	-0.0177	-0.010	773	< 0.0001
Quadratic Treatment Temperature (centered) - Reproduction	0.0004	-0.0041	0.005	773	0.85

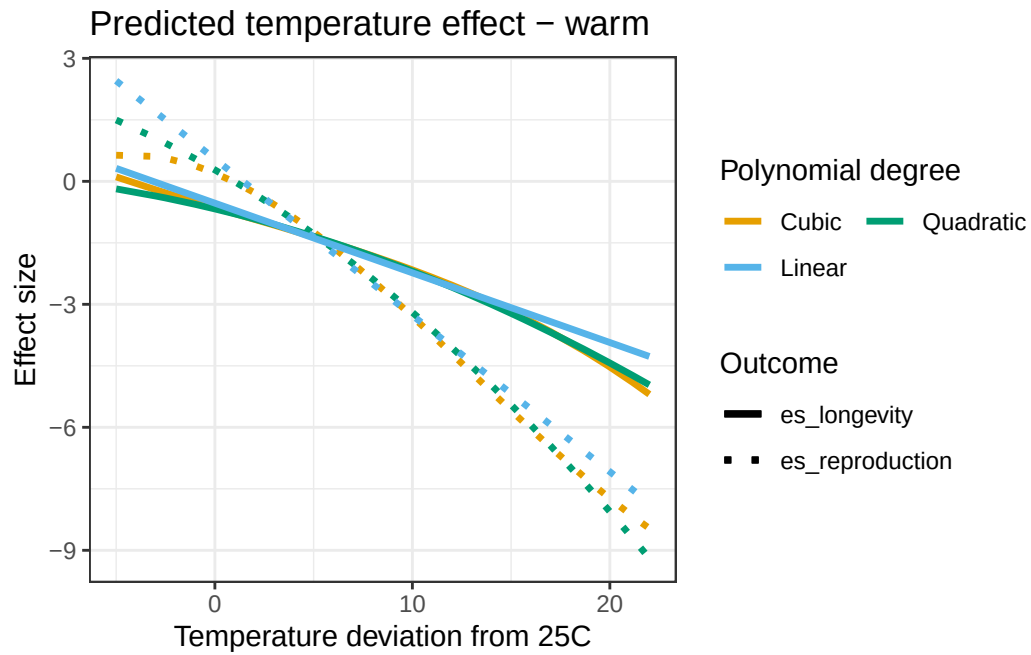
We found qualitatively similar correlations, after removing outliers (between-study correlation = -0.03, 95% CI: -0.24 to 0.19 within-study correlation = -0.15, 95% CI: -0.29 to 0)

Sensitivity analysis- chronic exposure only

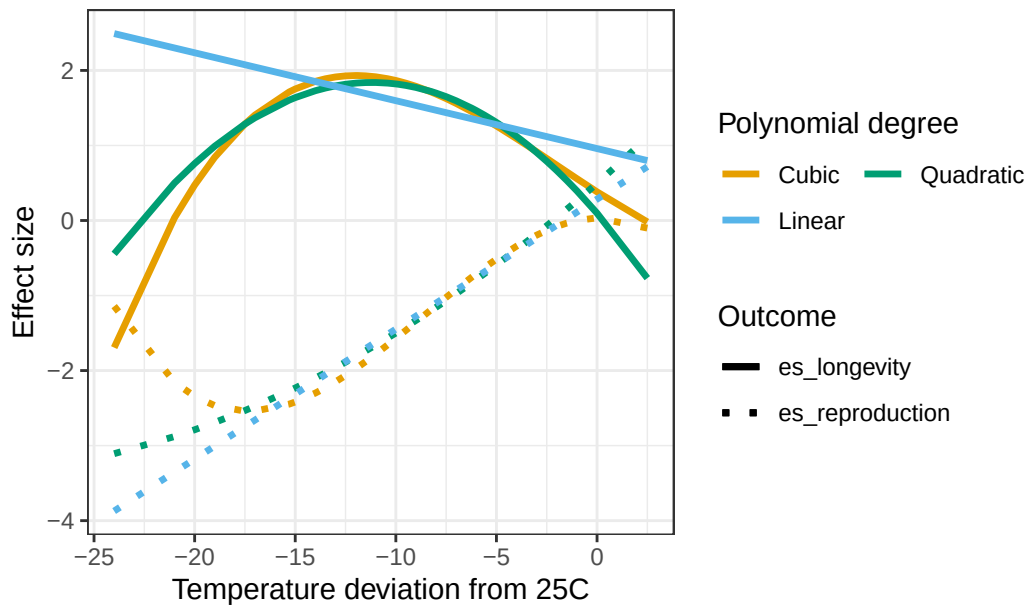
Warm subset

Cool subset

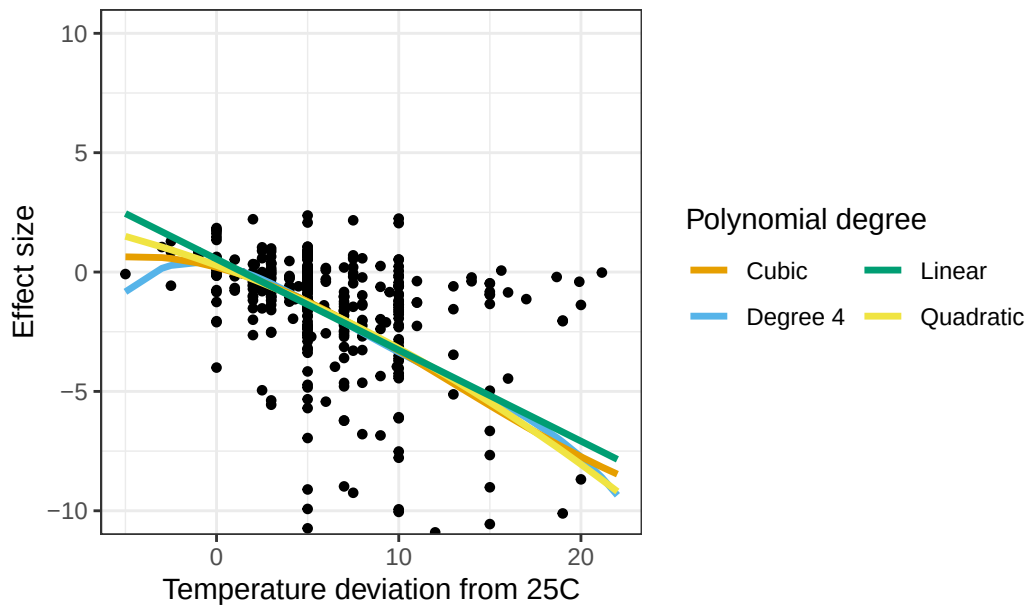
Higher-order polynomials



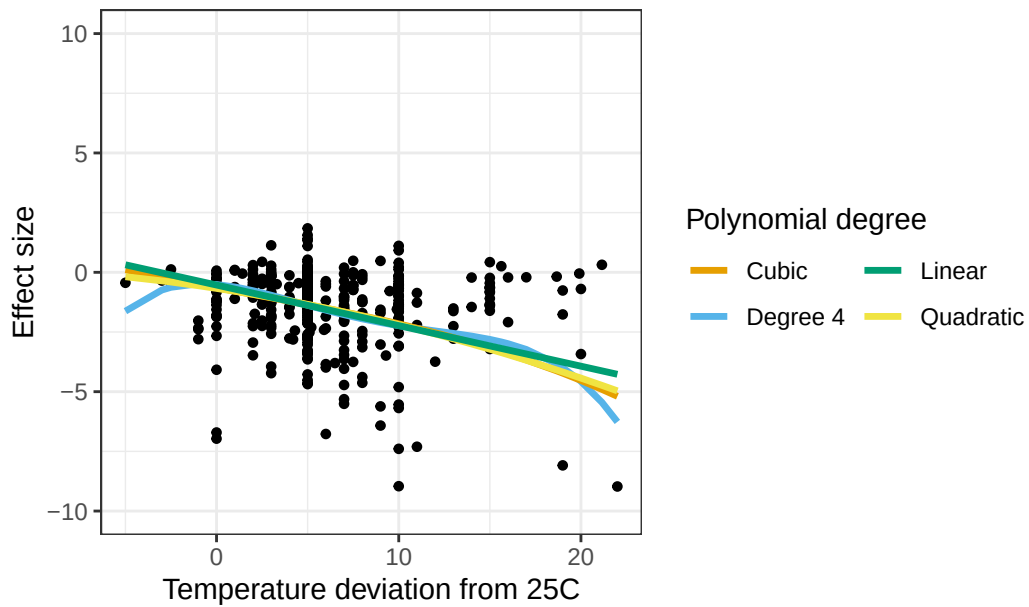
Predicted temperature effect – warm



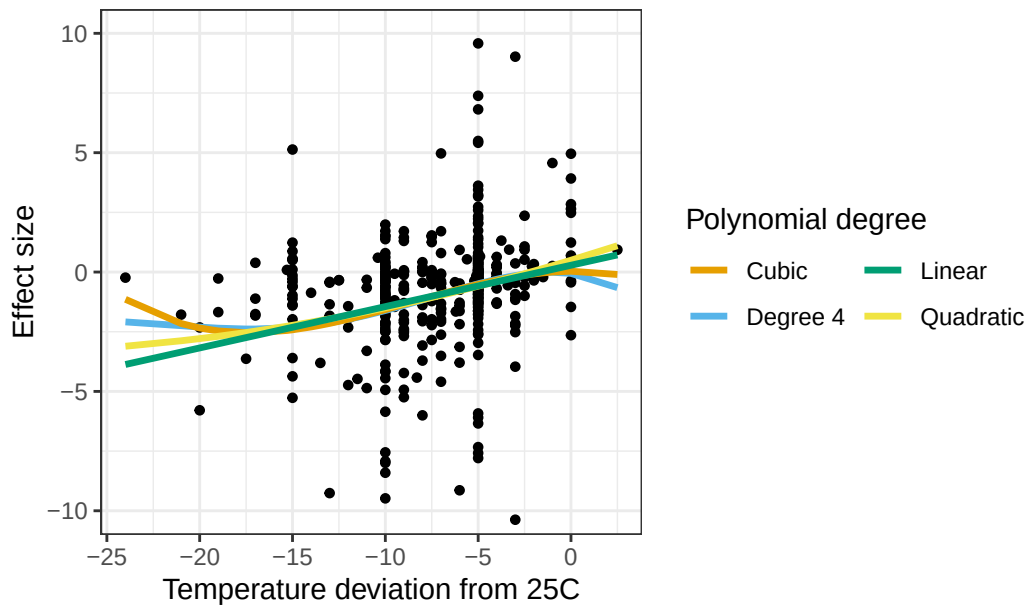
Predicted temperature effect – warm reproduction

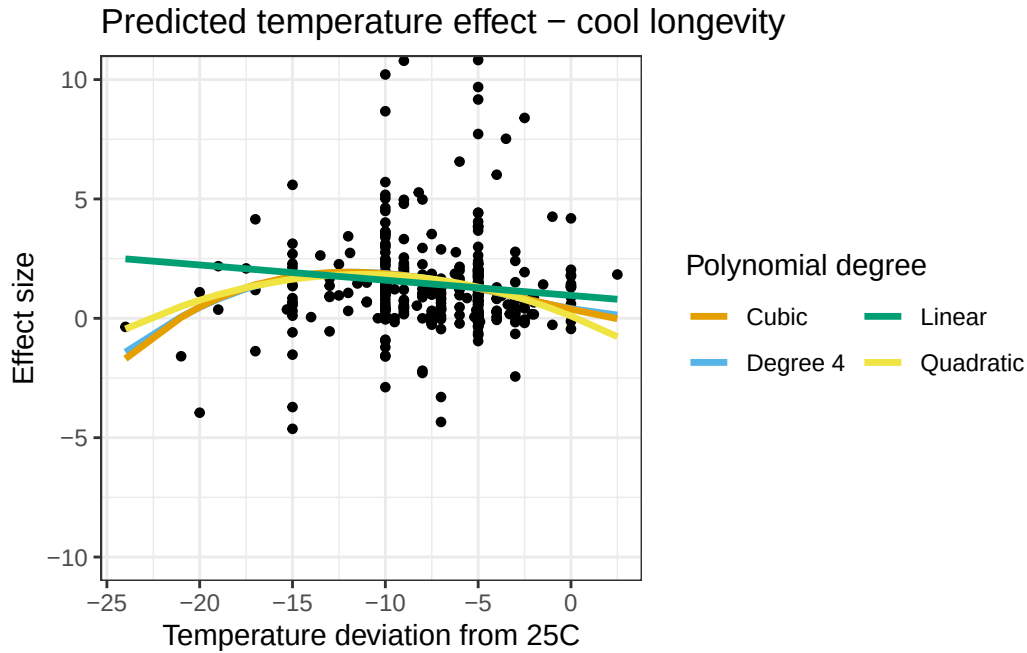


Predicted temperature effect – warm longevity



Predicted temperature effect – cool reproduction





Pest sub-analysis

Table 6: Sensitivity Analysis: Effects of temperature on mean effect size for longevity and reproduction when treatment temperatures are warmer than reference temperatures (with Aquatic species removed).

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Longevity (mean)	-0.57	-0.991	-0.145	153	< 0.01
Reproduction (mean)	-0.12	-0.196	-0.045	705	< 0.01
Linear Treatment Temperature (centered) - Longevity	-0.43	-0.807	-0.059	312	0.02
Linear Treatment Temperature (centered) - Reproduction	0.62	0.031	1.217	312	0.04
Quadratic Treatment Temperature (centered) - Longevity	-0.17	-0.207	-0.138	706	< 0.0001
Quadratic Treatment Temperature (centered) - Reproduction	-0.39	-0.449	-0.324	706	< 0.0001

Table 7: Sensitivity Analysis: Effects of temperature on mean effect size for longevity and reproduction when treatment temperatures are cooler than reference temperatures (with Aquatic species removed).

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Longevity (mean)	0.2861	-0.1055	0.6777	144	0.15
Reproduction (mean)	-0.2841	-0.3775	-0.1908	713	< 0.0001
Linear Treatment Temperature (centered) - Longevity	-0.3296	-0.3996	-0.2595	714	< 0.0001
Linear Treatment Temperature (centered) - Reproduction	0.1551	0.0775	0.2327	714	< 0.001
Quadratic Treatment Temperature (centered) - Longevity	-0.0148	-0.0193	-0.0103	714	< 0.0001
Quadratic Treatment Temperature (centered) - Reproduction	-0.0002	-0.0053	0.0049	714	0.94

Table 8: Sensitivity Analysis: Effects of temperature on mean effect size for longevity and reproduction when treatment temperatures are warmer than reference temperatures (with assumed rearing temp studies removed).

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Longevity (mean)	-0.47	-0.837	-0.11	123	0.01
Reproduction (mean)	1.08	0.516	1.64	250	< 0.001
Linear Treatment Temperature (centered) - Longevity	-0.47	-0.835	-0.11	250	0.01
Linear Treatment Temperature (centered) - Reproduction	0.61	-0.047	1.26	250	0.07
Quadratic Treatment Temperature (centered) - Longevity	-0.16	-0.197	-0.12	580	< 0.0001
Quadratic Treatment Temperature (centered) - Reproduction	-0.41	-0.478	-0.34	580	< 0.0001

Table 9: Sensitivity Analysis: Effects of temperature on mean effect size for longevity and reproduction when treatment temperatures are cooler than reference temperatures (with assumed rearing temp studies removed).

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Longevity (mean)	0.2601	-0.139	0.6588	108	0.2
Reproduction (mean)	-0.2608	-0.351	-0.1703	549	< 0.0001
Linear Treatment Temperature (centered) - Longevity	-0.3019	-0.368	-0.2360	550	< 0.0001

Parameters	Est.	L 95% CI	U 95% CI	df	p-value
Linear Treatment Temperature (centered) - Reproduction	0.1625	0.065	0.2604	550	< 0.01
Quadratic Treatment Temperature (centered) - Longevity	-0.0134	-0.018	-0.0091	550	< 0.0001
Quadratic Treatment Temperature (centered) - Reproduction	-0.0004	-0.007	0.0062	550	0.9